



UKUDLA Data Science Certificate

Data Management

Markus Mößler
University of Hohenheim



UKUDLA
African German Centre
for Sustainable and Resilient Food
Systems and Applied Agricultural
and Food Data Science



by decision of the
German Bundestag

Table of contents

1. Motivation
2. Concepts
3. Application
4. Key Message

Reproducible analysis can reproduce a mistake

The workflow runs

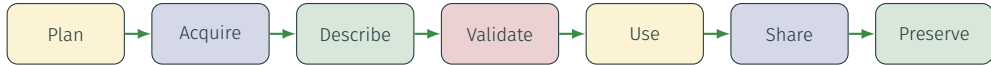
- code is versioned;
- packages are restored;
- the container starts; and
- the report is rendered.

But the data may be misunderstood

- a measurement uses the wrong unit;
- missing values are read as zero;
- flags are ignored; or
- a revised source replaces an earlier file.

Data management makes the meaning, origin, quality, organization, and permitted use of data explicit.

Data move through a lifecycle



- Decisions at one stage affect what is possible at later stages.
- Documentation and provenance connect the stages.
- Validation is repeated when data, requirements, or uses change.

This session moves from **motivation**, through the essential **concepts**, to their responsible **application** in a data workflow.

What does one row represent?

Dataset grain

The grain states what a single observation represents.

Depending on the study, one row may represent one:

- sample or specimen;
- plot or observational unit;
- measurement occasion;
- experimental treatment; or
- reported entity and period.

Role	Example
Identifier	Sample, plot, farm or region ID
Time	Collection date, season or year
Measure	Observed value
Meaning	Variable or assay
Context	Unit, treatment, method or flag

A key is the smallest set of variables expected to identify one row uniquely.

Food-systems evidence has different structures

Study structures

- Laboratory: aliquots within samples or batches
- Experiment: treatments assigned to field units
- Observation: farms, plots or people measured without assignment
- Secondary: reported entities and periods

Common dimensions

- Cross-sectional, longitudinal or repeated measures
- Hierarchical units, blocks, sites and batches
- Points, polygons, transects or raster cells
- Identifiers, protocols and boundaries may change

Structure determines valid keys, comparisons, summaries, and integration choices. A file format alone does not describe that structure.

Metadata and provenance answer different questions

Source metadata

- definitions;
- methods;
- classifications;
- code lists; and
- quality flags.

Data dictionary

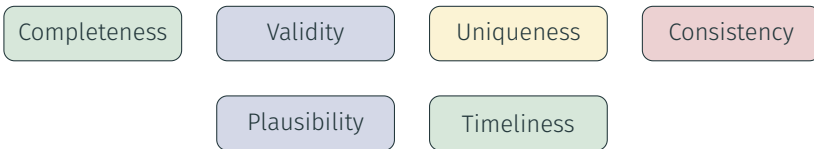
- variable names;
- meanings;
- types;
- units; and
- allowed values.

Provenance record

- collector or provider and dataset;
- protocol/version and collection/access date;
- license;
- retrieval method; and
- derived artifacts.

Metadata explain data; provenance explains their history.

Data quality means fitness for purpose



- Quality requirements depend on the research question and intended use.
- Accuracy often cannot be established from a file alone.
- Field notes, laboratory flags and provider notes are part of the evidence.
- A quality concern should be reported before it is corrected or excluded.

Application: validate before transforming

Structural checks

- expected columns and types;
- row count and temporal coverage;
- uniqueness of the proposed key;
- duplicate records; and
- allowed codes and units.

Semantic checks

- missingness patterns;
- impossible or suspicious values;
- consistency with definitions;
- agreement with metadata; and
- unresolved quality flags.

Validation should reveal assumptions and anomalies. It should not silently “fix” the evidence.

Organize the project by data role

```
analysis-project/  
|-- data-raw/  
|-- data-interim/  
|-- data-processed/  
|-- metadata/  
|-- scripts/  
`-- reports/
```

- Preserve raw inputs unchanged.
- Create derived data with code.
- Keep dictionaries, licenses and provenance near the project.
- Use stable names, not `final-v2-revised.csv`.
- Decide deliberately which data belong in Git.

Raw, interim and processed describe roles in a workflow—not levels of correctness.

Manage the project data responsibly

Before storing or sharing

- Check consent, license and terms of use.
- Record citation and attribution.
- Confirm whether redistribution is permitted.
- Protect personal, confidential and location-sensitive data.

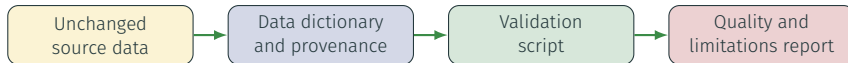
FAIR principles

- Findable
- Accessible
- Interoperable
- Reusable

FAIR does not necessarily mean open.

Farm, household, location and biological data can remain sensitive even after direct names are removed.

Key message: make data inspectable before analysis



1. State the grain and expected key.
2. Interpret variables, units, codes and flags.
3. Document source, meaning, license and history.
4. Validate structure and meaning without changing the raw file.
5. Record unresolved concerns for the next workflow stage.

The next session asks how to obtain and integrate multiple sources.

Supported by

South African government and the National Research Foundation



science, technology
& innovation

Department:
Science, Technology and Innovation
REPUBLIC OF SOUTH AFRICA



National
Research
Foundation

German government and the German Academic Exchange Service

Supported by:



Federal Foreign Office

With funding from the:



Federal Ministry
of Research, Technology
and Space

Supported by:



Federal Ministry
of Agriculture, Food
and Regional Identity

by decision of the
German Bundestag

Project Manager:



Federal Office
for Agriculture and Food



Deutscher Akademischer Austauschdienst
German Academic Exchange Service